

VTU-EASY.COM

Your Smart Study Partner

MODULE 5

Curve Fitting, Correlation & Regression

Numerical Methods | Engineering Mathematics

Topics Covered

Principles of Least Squares | Curve Fitting ($y = ax+b$, $y = ax^2+bx+c$, $y = ax^b$)
Correlation | Coefficient of Correlation | Lines of Regression
Angle Between Regression Lines | Standard Error of Estimate |
Rank Correlation

Prepared for VTU Students | www.vtu-easy.com

TABLE OF CONTENTS

vtu-easy.com

- [1. Introduction to Curve Fitting](#)
- [2. Principles of Least Squares](#)
 - [2.1 Definition and Concept](#)
 - [2.2 Mathematical Derivation](#)
- [3. Fitting a Straight Line: \$y = ax + b\$](#)
 - [3.1 Normal Equations](#)
 - [3.2 Alternate Form: \$y = a + bx\$](#)
 - [3.3 Solved Problems](#)
- [4. Fitting a Second Degree Parabola: \$y = ax^2 + bx + c\$](#)
 - [4.1 Normal Equations](#)
 - [4.2 Alternate Form: \$y = a + bx + cx^2\$](#)
 - [4.3 Solved Problems](#)
- [5. Fitting a Curve of the Form \$y = ax^b\$](#)
 - [5.1 Linearization Method \(Logarithmic Transformation\)](#)
 - [5.2 Normal Equations](#)
 - [5.3 Solved Problems](#)
- [6. Correlation and Coefficient of Correlation](#)
 - [6.1 Definition of Correlation](#)
 - [6.2 Types of Correlation](#)
 - [6.3 Karl Pearson's Coefficient of Correlation](#)
 - [6.4 Properties of Correlation Coefficient](#)
 - [6.5 Solved Problems on Correlation](#)
- [7. Lines of Regression](#)
 - [7.1 Definition and Types](#)
 - [7.2 Regression Line of \$y\$ on \$x\$](#)
 - [7.3 Regression Line of \$x\$ on \$y\$](#)
 - [7.4 Relationship Between \$r\$, \$b_{yx}\$, \$b_{xy}\$](#)
 - [7.5 Finding \$r\$ from Regression Lines](#)
 - [7.6 Solved Problems on Regression](#)
- [8. Angle Between Regression Lines](#)
 - [8.1 Formula and Derivation](#)
 - [8.2 Solved Problems](#)
- [9. Standard Error of Estimate](#)
- [10. Rank Correlation \(Spearman's\)](#)
 - [10.1 Definition and Formula](#)
 - [10.2 Type 1: When Ranks are Given](#)
 - [10.3 Type 2: When Ranks are NOT Given](#)
 - [10.4 Solved Problems](#)
- [11. Important Formulas Summary](#)

1. INTRODUCTION TO CURVE FITTING

vtu-easy.com

Curve Fitting is one of the most important techniques in Numerical Methods and Applied Mathematics. It is the process of finding a mathematical relation (equation) that best fits a given set of data points. The equation obtained in this process is called the Best Fitting Equation or Fitting Equation.

Definition: Curve Fitting is the method of finding a relation for a given data such that the derived equation fits the data in the best possible way, minimizing the discrepancy between observed and calculated values.

Why is Curve Fitting Important?

- In engineering and science, we frequently deal with experimental or observational data.
- We cannot always find an exact mathematical function passing through all data points.
- Curve fitting provides the best approximate mathematical equation to describe the data.
- It enables prediction of values at points not in the original dataset (interpolation/extrapolation).
- It is used extensively in regression analysis, machine learning, and statistical modelling.

Types of Curve Fitting Covered in This Module

Table 1.1: Types of Curve Fitting

Type	Equation Form	Method Used
Straight Line Fitting	$y = ax + b$ (or $y = a + bx$)	Least Squares
Parabola Fitting	$y = ax^2 + bx + c$ (or $y = a + bx + cx^2$)	Least Squares
Power Curve Fitting	$y = ax^b$	Linearization via Logarithm

NOTE: All curve fitting in this module uses the Principle of Least Squares, which guarantees the minimum total squared error between observed and fitted values.

KEY POINT: The primary goal of curve fitting is NOT to pass through every data point, but to find the equation that minimizes the total squared deviation (error) from all points.

2. PRINCIPLES OF LEAST SQUARES

vtu-easy.com

2.1 Definition and Concept

The Principle of Least Squares states that the best fitting curve is one for which the sum of the squares of the residuals (the differences between observed and calculated values) is a minimum.

Residual: For each data point (x_i, y_i) , if the curve gives a value $y(x_i)$, then the residual is $e_i = y_i - y(x_i)$.

$$S = \sum e_i^2 = \sum [y_i - f(x_i)]^2 \rightarrow \text{MINIMIZE}$$

The method minimizes S by choosing the parameters of the fitting equation. Squaring the residuals ensures that positive and negative deviations do not cancel each other, and larger deviations are penalized more heavily.

2.2 Mathematical Derivation

For a general fitting function $f(x)$ with parameters a, b (and possibly c), we minimize S with respect to each parameter by taking partial derivatives and setting them to zero:

$$\partial S / \partial a = 0, \quad \partial S / \partial b = 0, \quad \partial S / \partial c = 0$$

This produces a system of simultaneous linear equations called NORMAL EQUATIONS. Solving these normal equations gives the values of the unknown parameters a, b, c , which determine the best fitting curve.

Table 2.1: Parameters and Normal Equations

Fitting Equation	No. of Parameters	No. of Normal Equations
$y = ax + b$	2 (a, b)	2
$y = ax^2 + bx + c$	3 (a, b, c)	3
$y = ax^b$ (after log transform)	2 (A, B)	2

NOTE: The normal equations are always a system of linear equations in the unknown parameters, regardless of the original form of the fitting equation. This makes them easy to solve using standard linear algebra methods.

3. FITTING A STRAIGHT LINE: $y = ax + b$

vtu-easy.com

The most common and simplest form of curve fitting is fitting a straight line to a set of data points. The equation of the straight line is:

$$y = ax + b \dots(1)$$

3.1 Normal Equations

Using the Principle of Least Squares, we minimize $S = \sum(y_i - ax_i - b)^2$. Taking partial derivatives with respect to a and b and equating to zero, we get the following two normal equations:

$$\text{Normal Equation 1: } \sum y = a \sum x + nb$$

$$\text{Normal Equation 2: } \sum xy = a \sum x^2 + b \sum x$$

where n = number of data points. Solving these two equations simultaneously gives the values of a and b , and the best fitting straight line $y = ax + b$ is obtained.

3.2 Alternate Form: $y = a + bx$

Another common form of the straight line equation is $y = a + bx$, where a is the y -intercept and b is the slope. This gives the same normal equations, just written differently:

$$\text{Normal Equation 1: } \sum y = na + b \sum x$$

$$\text{Normal Equation 2: } \sum xy = a \sum x + b \sum x^2$$

NOTE: Both forms are equivalent. Solve the 2×2 linear system to find a and b . Either form ($y = ax + b$ or $y = a + bx$) may be used in exams — match the notation given in the question.

3.3 Solved Problems

SOLVED PROBLEM 1

Fit a straight line $y = ax + b$ for the following data:

Given Data ($n = 8$)

x	1	3	4	6	8	9	11	14
y	1	2	4	4	5	7	8	9

Solution: The normal equations are:

$$\begin{aligned}\Sigma y &= a\Sigma x + nb \dots(I) \\ \Sigma xy &= a\Sigma x^2 + b\Sigma x \dots(II)\end{aligned}$$

We compute the required sums using a tabular method:

Table 3.1: Computation Table

x	y	xy	x ²
1	1	1	1
3	2	6	9
4	4	16	16
6	4	24	36
8	5	40	64
9	7	63	81
11	8	88	121
14	9	126	196
$\Sigma x = 56$	$\Sigma y = 40$	$\Sigma xy = 364$	$\Sigma x^2 = 524$

Substituting in the normal equations:

$$\begin{aligned}(I): 40 &= 56a + 8b \\ (II): 364 &= 524a + 56b\end{aligned}$$

Solving this 2×2 system simultaneously:

Multiply equation (I) by 7: $280 = 392a + 56b$

Subtract from equation (II): $364 - 280 = (524 - 392)a$

$$84 = 132a \rightarrow a = 84/132 = 0.636$$

Substituting back in (I): $40 = 56(0.636) + 8b \rightarrow 40 = 35.616 + 8b \rightarrow b = 4.384/8 = 0.545$

$$\therefore \text{Best Fitting Line: } y = 0.636x + 0.545$$

SOLVED PROBLEM 2

Find the equation of a straight line for the following data and estimate y when x = 30:

Given Data (n = 5)

x	5	10	15	20	25
y	16	19	23	26	30

Solution: Using $y = a + bx$, the normal equations are:

$$\begin{aligned}\Sigma y &= na + b\Sigma x \\ \Sigma xy &= a\Sigma x + b\Sigma x^2\end{aligned}$$

Table 3.2: Computation Table

x	y	xy	x ²
5	16	80	25
10	19	190	100
15	23	345	225
20	26	520	400
25	30	750	625
$\Sigma x=75$	$\Sigma y=114$	$\Sigma xy=1885$	$\Sigma x^2=1375$

$$\begin{aligned}114 &= 5a + 75b \dots(I) \\ 1885 &= 75a + 1375b \dots(II)\end{aligned}$$

Multiply (I) by 15: $1710 = 75a + 1125b$

(II) – result: $175 = 250b \rightarrow b = 0.7$

From (I): $114 = 5a + 75(0.7) = 5a + 52.5 \rightarrow 5a = 61.5 \rightarrow a = 12.3$

$$\therefore y = 12.3 + 0.7x$$

When $x = 30$: $y = 12.3 + 0.7(30) = 12.3 + 21 = 33.3$

$$\therefore \text{Estimated value: } y = 33.3 \text{ when } x = 30$$

SOLVED PROBLEM 3 (YEAR DATA WITH SHIFTING)

Fit a straight line to the following production data and estimate for year 2006:

Given Data (n = 5)

Year (x)	1961	1971	1981	1991	2001
Production y (tons)	8	10	12	10	16

Solution: To simplify calculations, let $X = x - 1981$ (shifting origin to 1981):

Table 3.3: Computation with Shifted Variable

x (Year)	y	X = x-1981	XY	X ²
1961	8	-20	-160	400
1971	10	-10	-100	100
1981	12	0	0	0
1991	10	10	100	100
2001	16	20	320	400
	$\Sigma y=56$	$\Sigma X=0$	$\Sigma XY=160$	$\Sigma X^2=1000$

Since $\Sigma X = 0$, the normal equations simplify to:

$$56 = 5a + 0 \rightarrow a = 56/5 = 11.2$$

$$160 = 0 + 1000b \rightarrow b = 0.16$$

The line of fit: $y = 11.2 + 0.16X = 11.2 + 0.16(x - 1981)$

When $x = 2006$: $y = 11.2 + 0.16(2006 - 1981) = 11.2 + 0.16(25) = 11.2 + 4 = 15.2$

$\therefore$ Expected production in 2006 = 15.2 tons

4. FITTING A SECOND DEGREE PARABOLA

vtu-easy.com

When data shows a curved (non-linear) trend, a second degree (quadratic) parabola provides a better fit than a straight line. The equation is:

$$y = ax^2 + bx + c \dots(1)$$

4.1 Normal Equations for $y = ax^2 + bx + c$

Applying the Principle of Least Squares and differentiating $S = \sum(y_i - ax_i^2 - bx_i - c)^2$ with respect to a , b , c and setting to zero, we obtain three normal equations:

$$\text{Normal Equation I: } \sum y = a\sum x^2 + b\sum x + nc$$

$$\text{Normal Equation II: } \sum xy = a\sum x^3 + b\sum x^2 + c\sum x$$

$$\text{Normal Equation III: } \sum x^2y = a\sum x^4 + b\sum x^3 + c\sum x^2$$

Here n = number of data points. Solving these 3 equations simultaneously gives a , b , c .

4.2 Alternate Form: $y = a + bx + cx^2$

The alternate (and more common in VTU) form is $y = a + bx + cx^2$. The normal equations are:

$$\text{Normal Equation I: } \sum y = na + b\sum x + c\sum x^2$$

$$\text{Normal Equation II: } \sum xy = a\sum x + b\sum x^2 + c\sum x^3$$

$$\text{Normal Equation III: } \sum x^2y = a\sum x^2 + b\sum x^3 + c\sum x^4$$

NOTE: The sums needed for a parabola are: $\sum x$, $\sum y$, $\sum x^2$, $\sum x^3$, $\sum x^4$, $\sum xy$, $\sum x^2y$. Prepare a 7-column table for calculations.

4.3 Solved Problems

SOLVED PROBLEM 1 — PARABOLA

Fit a best fitting parabola $y = ax^2 + bx + c$ for the following data and estimate y at $x = 6$:

Given Data ($n = 5$)

x	1	2	3	4	5
y	10	12	13	16	19

Solution: The normal equations are (with $y = ax^2 + bx + c$):

$$\begin{aligned}\Sigma y &= a\Sigma x^2 + b\Sigma x + nc \dots(I) \\ \Sigma xy &= a\Sigma x^3 + b\Sigma x^2 + c\Sigma x \dots(II) \\ \Sigma x^2y &= a\Sigma x^4 + b\Sigma x^3 + c\Sigma x^2 \dots(III)\end{aligned}$$

Table 4.1: Full Computation Table

x	y	x ²	x ³	x ⁴	xy	x ² y
1	10	1	1	1	10	10
2	12	4	8	16	24	48
3	13	9	27	81	39	117
4	16	16	64	256	64	256
5	19	25	125	625	95	475
Σ15	Σ70	Σ55	Σ225	Σ979	Σ232	Σ906

Substituting sums into normal equations:

$$\begin{aligned}(I): 70 &= 55a + 15b + 5c \\ (II): 232 &= 225a + 55b + 15c \\ (III): 906 &= 979a + 225b + 55c\end{aligned}$$

Solving the 3×3 linear system (by elimination or Cramer's rule):

$$a \approx 0.29, \quad b \approx 0.49, \quad c \approx 9.4$$

$$\therefore \text{Best Fitting Parabola: } y = 0.29x^2 + 0.49x + 9.4$$

$$\text{When } x = 6: y = 0.29(36) + 0.49(6) + 9.4 = 10.44 + 2.94 + 9.4 = 22.78$$

$$\therefore \text{Estimated value at } x = 6: y \approx 22.78$$

SOLVED PROBLEM 2 — PARABOLA (ALTERNATE FORM)

Fit a parabola $y = a + bx + cx^2$ for the following data:

Given Data (n = 5)

x	0	1	2	3	4
y	1	1.8	1.3	2.5	2.3

Normal equations for $y = a + bx + cx^2$:

$$\begin{aligned}\Sigma y &= na + b\Sigma x + c\Sigma x^2 \dots(I) \\ \Sigma xy &= a\Sigma x + b\Sigma x^2 + c\Sigma x^3 \dots(II)\end{aligned}$$

$$\Sigma x^2 y = a \Sigma x^2 + b \Sigma x^3 + c \Sigma x^4 \dots (III)$$

Table 4.2: Computation Table

x	y	x ²	x ³	x ⁴	xy	x ² y
0	1	0	0	0	0	0
1	1.8	1	1	1	1.8	1.8
2	1.3	4	8	16	2.6	5.2
3	2.5	9	27	81	7.5	22.5
4	2.3	16	64	256	9.2	36.8
Σ10	8.9	30	100	354	21.1	66.3

$$8.9 = 5a + 10b + 30c \dots (I)$$

$$21.1 = 10a + 30b + 100c \dots (II)$$

$$66.3 = 30a + 100b + 354c \dots (III)$$

Solving: $a \approx 1.0771$, $b \approx 0.4157$, $c \approx -0.0214$

$$\therefore y = 1.0771 + 0.4157x - 0.0214x^2$$

5. FITTING A CURVE OF THE FORM $y = ax^b$

vtu-easy.com

The equation $y = ax^b$ is a power curve (also called a geometric curve). It cannot be fitted directly using least squares in its original form because it is non-linear. However, it can be LINEARIZED by taking logarithms on both sides.

5.1 Linearization Method (Logarithmic Transformation)

Starting with $y = ax^b$, we take natural log (ln) of both sides:

$$\log y = \log a + b \log x$$

Now substitute: $Y = \log y$, $X = \log x$, $A = \log a$, $B = b$

The equation becomes: $Y = A + BX$, which is a straight line in Y and X !

Table 5.1: Variable Transformation

Original Variable	Transformed Variable	Relationship
y	$Y = \log y$ (or $\ln y$)	Y is the dependent variable in the linear system
x	$X = \log x$ (or $\ln x$)	X is the independent variable in the linear system
a	$A = \log a \rightarrow a = e^A$	A is the intercept; recover a by $a = e^A$
b	$B = b$	B is directly the slope of the fitted line

5.2 Normal Equations

Since $Y = A + BX$ is a straight line, the normal equations are (for n data points):

$$\text{Normal Equation I: } \sum Y = nA + B \sum X$$

$$\text{Normal Equation II: } \sum XY = A \sum X + B \sum X^2$$

Solving these two equations gives A and B . Then:

$$a = e^A \text{ and } b = B$$

NOTE: Always use natural logarithm (ln) consistently throughout the problem. If you use log base 10 in the table, you must also use log base 10 in the formula $a = 10^A$.

5.3 Solved Problems

SOLVED PROBLEM 1

Fit a best fitting equation of the form $y = ax^b$ for the following data and calculate y when $x = 3.5$:

Given Data ($n = 6$)

x	1	2	3	4	5	6
y	2.98	4.26	5.21	6.1	6.8	7.5

Solution: Taking natural log, let $Y = \ln y$ and $X = \ln x$:

Table 5.2: Log-transformed Computation Table

x	y	Y = $\ln y$	X = $\ln x$	X^2	XY
1	2.98	1.0912	0	0	0
2	4.26	1.4493	0.6931	0.4803	1.0045
3	5.21	1.6506	1.0986	1.2069	1.8133
4	6.1	1.8083	1.3862	1.9215	2.5066
5	6.8	1.9169	1.6094	2.5901	3.0850
6	7.5	2.0149	1.7917	3.2101	3.6100
Σ		$\Sigma Y = 9.9312$	$\Sigma X = 6.5790$	$\Sigma X^2 = 9.4089$	$\Sigma XY = 12.0194$

Normal equations:

$$9.9312 = 6A + 6.5790B \dots(I)$$

$$12.0194 = 6.5790A + 9.4089B \dots(II)$$

Solving: $A = 1.0908$, $B = 0.5147$

Recovering original parameters:

$$a = e^A = e^{1.0908} = 2.9766$$

$$b = B = 0.5147$$

$$\therefore \text{The curve is: } y = 2.9766 \cdot x^{0.5147}$$

When $x = 3.5$: $y = 2.9766 \times (3.5)^{0.5147} = 2.9766 \times 1.9050 \approx 5.6722$

$$\therefore y = 5.6722 \text{ when } x = 3.5$$

SOLVED PROBLEM 2

Fit a least square geometric curve $y = ax^b$ for: x : 1, 2, 3, 4, 5 and y : 0.5, 2, 4.5, 8, 12.5

Table 5.3: Computation Table

x	y	X = ln x	Y = ln y	XY	X ²
1	0.5	0	-0.6931	0	0
2	2	0.6931	0.6931	0.4804	0.4804
3	4.5	1.0986	1.5041	1.6524	1.2069
4	8	1.3863	2.0794	2.8829	1.9218
5	12.5	1.6094	2.5257	4.0649	2.5902
Σ		4.7874	6.1092	9.0806	6.1993

$$6.1092 = 5A + 4.7874B$$

$$9.0806 = 4.7874A + 6.1993B$$

Solving: $A = -0.69315$, $B = 2$

$$a = e^{(-0.69315)} = 0.5$$

$\therefore y = 0.5x^2$ is the required curve

6. CORRELATION AND COEFFICIENT OF CORRELATION

vtu-easy.com

6.1 Definition of Correlation

Correlation refers to the co-variation (joint variation) of two independent magnitudes or variables. In statistics, correlation measures the degree to which two variables are related or vary together.

Formal Definition: If two variables x and y are related in such a way that a change in one variable corresponds to a change in the other variable, then the variables are said to be correlated.

6.2 Types of Correlation

Table 6.1: Types of Correlation

Type	Description	Coefficient Value
Positive Correlation	Both variables increase or decrease together	$r > 0$ ($0 < r \leq 1$)
Negative Correlation	One increases while the other decreases	$r < 0$ ($-1 \leq r < 0$)
Perfect Positive	Exact linear increase in both	$r = +1$
Perfect Negative	Exact linear decrease relationship	$r = -1$
No Correlation	No linear relationship	$r = 0$

6.3 Karl Pearson's Coefficient of Correlation

The numerical measure of correlation between two variables x and y is known as the Correlation Coefficient (also called Pearson's Coefficient of Correlation or Karl Pearson's Coefficient). It is denoted by r and defined as:

$$r = \Sigma XY / (\sqrt{\Sigma X^2} \cdot \sqrt{\Sigma Y^2})$$

where $X = x - \bar{x}$ and $Y = y - \bar{y}$ (deviations from mean)

and $\bar{x} = \Sigma x/n$, $\bar{y} = \Sigma y/n$ (arithmetic means)

The formula can also be written in standard deviation form:

$$b_{yx} = r \cdot (\sigma_y / \sigma_x) \text{ and } b_{xy} = r \cdot (\sigma_x / \sigma_y)$$

where $\sigma_x = \sqrt{(\Sigma X^2/n)}$ and $\sigma_y = \sqrt{(\Sigma Y^2/n)}$ are the standard deviations.

6.4 Properties of Correlation Coefficient

- The coefficient of correlation r is dimensionless (pure number).
- r always lies between -1 and $+1$, i.e., $-1 \leq r \leq 1$.
- If $r = +1$, x and y are perfectly positively correlated.
- If $r = -1$, x and y are perfectly negatively correlated.
- If $r = 0$, x and y are not linearly correlated (but may have non-linear relationship).
- r is symmetric: the correlation of x with y equals the correlation of y with x .
- r is independent of change of origin and scale.

6.5 Solved Problems on Correlation

SOLVED PROBLEM 1

Compute the coefficient of correlation for: $x: 1, 2, 3, 4, 5, 6, 7$ and $y: 9, 8, 10, 12, 11, 13, 14$

Solution: $n = 7$

$$\bar{x} = (1+2+3+4+5+6+7)/7 = 28/7 = 4$$

$$\bar{y} = (9+8+10+12+11+13+14)/7 = 77/7 = 11$$

Table 6.2: Computation for Correlation

x	y	$X = x - 4$	$Y = y - 11$	XY	X^2	Y^2
1	9	-3	-2	6	9	4
2	8	-2	-3	6	4	9
3	10	-1	-1	1	1	1
4	12	0	1	0	0	1
5	11	1	0	0	1	0
6	13	2	2	4	4	4
7	14	3	3	9	9	9
		$\Sigma X = 0$	$\Sigma Y = 0$	$\Sigma XY = 26$	$\Sigma X^2 = 28$	$\Sigma Y^2 = 28$

$$r = \Sigma XY / (\sqrt{\Sigma X^2} \cdot \sqrt{\Sigma Y^2}) = 26 / (\sqrt{28} \cdot \sqrt{28}) = 26/28 = 0.9285 \approx 0.93$$

KEY POINT: $r = 0.93$ indicates a very strong positive correlation between x and y .

SOLVED PROBLEM 2

The following table gives heights of fathers (x) and sons (y). Calculate correlation coefficient:

Heights Data (n = 8)

x	65	66	67	67	68	69	70	72
y	67	68	65	68	72	72	69	71

$$\bar{x} = (65+66+67+67+68+69+70+72)/8 = 544/8 = 68$$

$$\bar{y} = (67+68+65+68+72+72+69+71)/8 = 552/8 = 69$$

Table 6.3

x	y	X=x-68	Y=y-69	XY	X ²	Y ²
65	67	-3	-2	6	9	4
66	68	-2	-1	2	4	1
67	65	-1	-4	4	1	16
67	68	-1	-1	1	1	1
68	72	0	3	0	0	9
69	72	1	3	3	1	9
70	69	2	0	0	4	0
72	71	4	2	8	16	4
				$\Sigma XY=24$	$\Sigma X^2=36$	$\Sigma Y^2=44$

$$r = 24 / (\sqrt{36} \cdot \sqrt{44}) = 24 / (6 \times 6.633) = 24/39.799 = 0.603$$

Standard deviations: $\sigma_x = \sqrt{(36/8)} = 2.1213$, $\sigma_y = \sqrt{(44/8)} = 2.3452$

7. LINES OF REGRESSION

vtu-easy.com

7.1 Definition and Types

Regression is the statistical process of estimating one variable (dependent) in terms of another variable (independent). When two variables x and y are correlated, the best fitting straight line in the least square sense gives a reasonably good relationship between x and y . This best-fitting line is called the Regression Line.

Table 7.1: Two Lines of Regression

Regression Line	Equation Form	Independent Variable	Use
y on x	$y - \bar{y} = byx(x - \bar{x})$	x	Predict y for given x
x on y	$x - \bar{x} = bxy(y - \bar{y})$	y	Predict x for given y

7.2 Regression Line of y on x

The regression line of y on x gives the value of y for a given value of x. Its equation is:

$$y - \bar{y} = byx (x - \bar{x})$$

where byx is called the Regression Coefficient of y on x, defined as:

$$byx = r \cdot (\sigma_y / \sigma_x)$$

Here r = correlation coefficient, σ_y = standard deviation of y, σ_x = standard deviation of x.

7.3 Regression Line of x on y

The regression line of x on y gives the value of x for a given value of y. Its equation is:

$$x - \bar{x} = bxy (y - \bar{y})$$

where bxy is the Regression Coefficient of x on y:

$$bxy = r \cdot (\sigma_x / \sigma_y)$$

7.4 Relationship Between r, byx , bxy

The correlation coefficient r is the geometric mean of the two regression coefficients:

$$r = \pm \sqrt{(byx \times bxy)}$$

KEY POINT: The sign of r is the same as the common sign of byx and bxy . If both regression coefficients are positive, r is positive; if both are negative, r is negative. If byx and bxy have opposite signs, something is wrong in calculation.

NOTE: Both regression lines pass through the point $(\bar{x}, \bar{y})$. This is an important property used to find the means when only regression line equations are given.

7.5 Finding r from Regression Lines

Given two regression line equations, we can find r by:

Step 1: Identify lines — Determine which equation is y on x and which is x on y .

Step 2: Extract coefficients — Extract byx (slope of y on x line) and bxy (slope of x on y line).

Step 3: Find r — Compute $r = \pm \sqrt{(byx \times bxy)}$, choosing sign same as byx and bxy .

Step 4: Find means — Solve the two equations simultaneously (since both pass through $(\bar{x}, \bar{y})$).

7.6 Solved Problems on Regression

SOLVED PROBLEM 1 — COMPUTING REGRESSION LINES

Compute the coefficient of correlation and regression lines for: $x: 1,2,3,4,5,6,7$ and $y: 9,8,10,12,11,13,14$

From the correlation problem: $n=7$, $\bar{x}=4$, $\bar{y}=11$, $\Sigma XY=26$, $\Sigma X^2=28$, $\Sigma Y^2=28$, $r=0.93$, $\sigma_x=2$, $\sigma_y=2$

Regression Line of y on x :

$$y - \bar{y} = byx(x - \bar{x}) \text{ where } byx = r \cdot (\sigma_y / \sigma_x) = 0.93 \cdot (2/2) = 0.93$$

$$y - 11 = 0.93(x - 4)$$

$$y = 0.93x - 3.72 + 11$$

$$\text{Regression Line (y on x): } y = 0.93x + 7.28$$

Regression Line of x on y :

$$x - \bar{x} = bxy(y - \bar{y}) \text{ where } bxy = r \cdot (\sigma_x / \sigma_y) = 0.93 \cdot (2/2) = 0.93$$

$$x - 4 = 0.93(y - 11)$$

$$x = 0.93y - 10.23 + 4$$

$$\text{Regression Line (x on y): } x = 0.93y - 6.23$$

SOLVED PROBLEM 2 — FATHERS AND SONS HEIGHTS

From Problem 2 of correlation: $n=8$, $\bar{x}=68$, $\bar{y}=69$, $r=0.603$, $\sigma_x=2.1213$, $\sigma_y=2.3452$

Regression Line of y on x:

$$b_{yx} = r \cdot (\sigma_y / \sigma_x) = 0.603 \cdot (2.3452 / 2.1213) = 0.603 \times 1.1055 = 0.6664$$

$$y - 69 = 0.6664(x - 68)$$

$$y = 0.6664x - 45.315 + 69$$

$$y = 0.6664x + 23.685$$

Regression Line of x on y:

$$b_{xy} = r \cdot (\sigma_x / \sigma_y) = 0.603 \cdot (2.1213 / 2.3452) = 0.603 \times 0.9045 = 0.5454$$

$$x - 68 = 0.5454(y - 69)$$

$$x = 0.5454y + 30.366$$

SOLVED PROBLEM 3 — FINDING $\bar{X}$, $\bar{Y}$, R FROM REGRESSION LINES

With usual notations, compute $\bar{x}$, $\bar{y}$ and r from the following regression lines: $2x + 3y + 1 = 0$ and $x + 6y - 4 = 0$

Solution: Since both regression lines pass through $(\bar{x}, \bar{y})$, substitute $\bar{x}$ and $\bar{y}$:

$$2\bar{x} + 3\bar{y} + 1 = 0 \rightarrow 2\bar{x} + 3\bar{y} = -1 \quad \dots(I)$$

$$\bar{x} + 6\bar{y} - 4 = 0 \rightarrow \bar{x} + 6\bar{y} = 4 \quad \dots(II)$$

$$\text{Multiply (II) by 2: } 2\bar{x} + 12\bar{y} = 8 \quad \dots(III)$$

$$(III) - (I): 9\bar{y} = 9 \rightarrow \bar{y} = 1; \text{ from (II): } \bar{x} = 4 - 6 = -2$$

$$\therefore \bar{x} = -2, \bar{y} = 1$$

To find r : Assume $2x + 3y + 1 = 0$ is the line of regression x on y :

$$x = -3y/2 - 1/2 \rightarrow b_{xy} = -3/2$$

Assume $x + 6y - 4 = 0$ is the line of regression y on x :

$$y = -x/6 + 2/3 \rightarrow b_{yx} = -1/6$$

$$\text{Check: } r^2 = b_{xy} \times b_{yx} = (-3/2)(-1/6) = 1/4 \rightarrow r = \pm 1/2$$

Since $r = 0.5 < 1$, the assumption is correct.

$$\therefore r = -0.5 \text{ (negative since both coefficients are negative)}$$

SOLVED PROBLEM 4 — HUSBAND-WIFE AGE EXAMPLE

The following data gives age of husband (x) and wife (y). Form 2 regression lines and find husband age when wife is 16.

Given Data (n = 10)

x	36	23	27	28	28	29	30	31	33	35
y	29	18	20	22	27	21	29	27	29	28

$$\bar{x} = 300/10 = 30, \quad \bar{y} = 250/10 = 25$$

Table 7.2

x	y	X=x-30	Y=y-25	XY	X ²	Y ²
36	29	6	4	24	36	16
23	18	-7	-7	49	49	49
27	20	-3	-5	15	9	25
28	22	-2	-3	6	4	9
28	27	-2	2	-4	4	4
29	21	-1	-4	4	1	16
30	29	0	4	0	0	16
31	27	1	2	2	1	4
33	29	3	4	12	9	16
35	28	5	3	15	25	9
				$\Sigma XY = 123$	$\Sigma X^2 = 138$	$\Sigma Y^2 = 164$

$$r = 123 / (\sqrt{138} \cdot \sqrt{164}) = 123 / (11.747 \times 12.806) = 123/150.44 = 0.8176 \approx 0.82$$

$$\sigma_x = \sqrt{(138/10)} = 3.714, \quad \sigma_y = \sqrt{(164/10)} = 4.0496$$

$$b_{yx} = 0.82 \times (4.0496/3.714) = 0.82 \times 1.0904 = 0.894$$

$$y - 25 = 0.894(x - 30) \rightarrow y = 0.894x - 1.82$$

$$b_{xy} = 0.82 \times (3.714/4.0496) = 0.82 \times 0.917 = 0.7520$$

$$x - 30 = 0.7520(y - 25) \rightarrow x = 0.7520y + 11.2$$

$$\therefore \text{Regression lines: } y = 0.894x - 1.82 \text{ and } x = 0.752y + 11.2$$

$$\text{When wife's age } y = 16: x = 0.752(16) + 11.2 = 12.032 + 11.2 = 23.232 \approx 23$$

$$\therefore \text{Husband's age when wife is 16} = 23 \text{ years}$$

8. ANGLE BETWEEN REGRESSION LINES

vtU-easy.com

8.1 Formula and Derivation

The two lines of regression are not the same unless there is perfect correlation ($r = \pm 1$). The angle θ between the two regression lines gives a geometric interpretation of the correlation between the variables.

If m_1 and m_2 are the slopes of two lines $y = m_1x + c_1$ and $y = m_2x + c_2$, the acute angle θ between them is:

$$\tan \theta = |m_2 - m_1| / |1 + m_1m_2|$$

The slopes of the two regression lines are:

Regression line of y on x : $m_1 = r \cdot (\sigma_y / \sigma_x)$

Regression line of x on y written as y on x : $m_2 = (1/r) \cdot (\sigma_y / \sigma_x)$

Substituting and simplifying:

$$\tan \theta = (\sigma_x \cdot \sigma_y / (\sigma_x^2 + \sigma_y^2)) \cdot ((1 - r^2) / r)$$

NOTE: Special cases: (1) If $r = 0$, $\tan \theta \rightarrow \infty$, so $\theta = 90^\circ$ — the lines are perpendicular when there is no correlation. (2) If $r = \pm 1$, $\tan \theta = 0$, so $\theta = 0^\circ$ — both lines coincide for perfect correlation.

8.2 Solved Problems

SOLVED PROBLEM 1

Show that if θ is the angle between the lines of regression, then $\tan \theta = (\sigma_x \cdot \sigma_y / (\sigma_x^2 + \sigma_y^2)) \cdot ((1 - r^2) / r)$

Proof: The regression line of y on x is: $y - \bar{y} = r \cdot (\sigma_y / \sigma_x) \cdot (x - \bar{x})$

This can be written in slope-intercept form with slope $m_1 = r \cdot (\sigma_y / \sigma_x)$

The regression line of x on y : $x - \bar{x} = r \cdot (\sigma_x / \sigma_y) \cdot (y - \bar{y})$

Rewriting as y in terms of x : $y - \bar{y} = (1/r) \cdot (\sigma_y / \sigma_x) \cdot (x - \bar{x})$

This has slope $m_2 = (1/r) \cdot (\sigma_y / \sigma_x)$

Now, $\tan \theta = (m_2 - m_1) / (1 + m_1m_2)$

$$m_2 - m_1 = (\sigma_y/\sigma_x) \cdot [(1/r) - r] = (\sigma_y/\sigma_x) \cdot (1 - r^2)/r$$

$$m_1 \cdot m_2 = r \cdot (\sigma_y/\sigma_x) \times (1/r) \cdot (\sigma_y/\sigma_x) = (\sigma_y/\sigma_x)^2 = \sigma_y^2/\sigma_x^2$$

$$1 + m_1 m_2 = 1 + \sigma_y^2/\sigma_x^2 = (\sigma_x^2 + \sigma_y^2)/\sigma_x^2$$

$$\text{Therefore: } \tan \theta = [(\sigma_y/\sigma_x) \cdot (1-r^2)/r] / [(\sigma_x^2 + \sigma_y^2)/\sigma_x^2]$$

$$= (\sigma_y \cdot \sigma_x \cdot (1-r^2)/r) / (\sigma_x^2 + \sigma_y^2)$$

$$\therefore \tan \theta = (\sigma_x \cdot \sigma_y / (\sigma_x^2 + \sigma_y^2)) \times ((1-r^2)/r) \quad [\text{Proved}]$$

SOLVED PROBLEM 2

In a bivariate distribution, $\sigma_x = \sigma_y$ and the acute angle between regression lines is $\tan^{-1}(3)$. Find r .

Given: $\sigma_x = \sigma_y$ and $\tan \theta = 3$

Using the formula with $\sigma_x = \sigma_y$:

$$\tan \theta = (\sigma_x \cdot \sigma_y / (\sigma_x^2 + \sigma_y^2)) \times ((1-r^2)/r)$$

$$= (\sigma_x^2 / 2\sigma_x^2) \times ((1-r^2)/r)$$

$$= (1/2) \times ((1-r^2)/r)$$

$$3 = (1/2) \cdot (1-r^2)/r \rightarrow 6r = 1 - r^2 \rightarrow r^2 + 6r - 1 = 0$$

$$r = [-6 \pm \sqrt{(36+4)}] / 2 = [-6 \pm \sqrt{40}] / 2$$

$$r = (-6 + 6.3246)/2 = 0.1623 \text{ or } r = -6.1623 < -1 \text{ (rejected)}$$

$$\therefore r = 0.1622$$

9. STANDARD ERROR OF ESTIMATE

vtu-easy.com

The Standard Error of Estimate is a measure of the accuracy of predictions made with the regression line. It measures the average scatter of the observed data points around the regression line.

For the regression line of y on x, the standard error of estimate (S_{yx}) is:

$$S_{yx} = \sigma_y \cdot \sqrt{1 - r^2}$$

For the regression line of x on y, the standard error of estimate (S_{xy}) is:

$$S_{xy} = \sigma_x \cdot \sqrt{1 - r^2}$$

NOTE: A smaller standard error means the regression line fits the data better. If $r = \pm 1$ (perfect correlation), the standard error = 0, meaning the line passes through all data points exactly.

Table 9.1: Standard Error vs Correlation

Value of r	Standard Error (S_{yx})	Interpretation
$r = \pm 1$	$S_{yx} = 0$	Perfect fit; line passes through all points
$r = 0$	$S_{yx} = \sigma_y$	Worst fit; no linear relationship
$0 < r < 1$	$0 < S_{yx} < \sigma_y$	Partial correlation; some scatter around line

KEY POINT: The standard error of estimate is always between 0 and σ_y (the standard deviation of y). A value near 0 indicates a good fit; a value near σ_y indicates a poor fit.

EXAMPLE

If for a dataset, $\sigma_x = 3$, $\sigma_y = 4$, and $r = 0.8$, find the standard errors of estimate.

$$S_{yx} = \sigma_y \cdot \sqrt{1 - r^2} = 4 \cdot \sqrt{1 - 0.64} = 4 \cdot \sqrt{0.36} = 4 \times 0.6 = 2.4$$

$$S_{xy} = \sigma_x \cdot \sqrt{1 - r^2} = 3 \cdot \sqrt{1 - 0.64} = 3 \times 0.6 = 1.8$$

10. RANK CORRELATION (SPEARMAN'S)

vtu-easy.com

10.1 Definition and Formula

Spearman's Rank Correlation is a non-parametric measure of correlation that uses the ranks of variables rather than their actual values. It is defined as the Pearson correlation coefficient between the rank variables.

Definition: If x and y are marks scored in two subjects, let R_x and R_y be the ranks, and $d = R_x - R_y$ be the differences of ranks. Then Spearman's rank correlation coefficient ρ (rho) is:

$$\rho = 1 - (6\sum d^2) / (n^3 - n)$$

where n = number of data pairs, $d = R_x - R_y$ = difference in ranks.

NOTE: ρ ranges between -1 and $+1$, just like Pearson's r . $\rho = +1$ means perfect agreement in rankings; $\rho = -1$ means perfect inverse ranking; $\rho = 0$ means no rank correlation.

10.2 Type 1: When Ranks are Given

When ranks are directly given, simply compute $d = R_x - R_y$ for each pair, then d^2 , and apply the formula.

SOLVED PROBLEM 1 — TYPE 1

The ranking of 10 students in two subjects A and B are given below. Calculate rank correlation coefficient.

Given Ranks ($n = 10$)

Student No.	1	2	3	4	5	6	7	8	9	10
Rank A (RA)	3	5	8	4	7	10	2	1	6	9
Rank B (RB)	6	4	9	8	1	2	3	10	5	7

Table 10.1: Rank Difference Table

RA	RB	d = RA-RB	d ²
3	6	-3	9
5	4	1	1
8	9	-1	1
4	8	-4	16
7	1	6	36
10	2	8	64
2	3	-1	1
1	10	-9	81
6	5	1	1
9	7	2	4
			$\Sigma d^2 = 214$

$$\rho = 1 - (6 \times 214) / (10^3 - 10) = 1 - 1284/990 = 1 - 1.2967 = -0.2967$$

$$\therefore \rho = -0.2967$$

SOLVED PROBLEM 2 — BEAUTY CONTEST

Ten competitors in a beauty contest are ranked by two judges A and B. Calculate rank correlation.

Given Data (n=10)

Competitor	1	2	3	4	5	6	7	8	9	10
Judge A (RA)	1	6	5	3	10	2	4	9	7	8
Judge B (RB)	6	4	9	8	1	2	3	10	5	7

Table 10.2

RA	RB	d	d ²
1	6	-5	25
6	4	2	4
5	9	-4	16
3	8	-5	25
10	1	9	81
2	2	0	0
4	3	1	1

RA	RB	d	d ²
9	10	-1	1
7	5	2	4
8	7	1	1
			$\Sigma d^2 = 158$

$$\rho = 1 - (6 \times 158) / (10^3 - 10) = 1 - 948 / 990 = 1 - 0.9576 = 0.0424$$

$$\therefore \rho = 0.0424$$

10.3 Type 2: When Ranks are NOT Given

When actual data (scores/values) are given instead of ranks:

Step 1: Rank each variable — Assign Rank 1 to the highest value, Rank 2 to next, etc. If values are equal (ties), assign the average rank to all tied values.

Step 2: Compute differences — Find $d = R_x - R_y$ for each pair.

Step 3: Apply formula — Compute $\rho = 1 - 6\Sigma d^2 / (n^3 - n)$.

10.4 Solved Problems (Type 2)

SOLVED PROBLEM 3 — SCORES IN ENGLISH AND MATHEMATICS

The scores for 10 students in English and Mathematics are given. Calculate rank correlation.

Given Data ($n = 10$)

English (x)	56	75	45	71	62	64	58	80	76	61
Maths (y)	66	70	40	60	65	56	59	77	67	63

Table 10.3: Ranking and Differences

Eng	Maths	R _x	R _y	d = R _x - R _y	d ²
56	66	9	4	5	25
75	70	3	2	1	1
45	40	10	10	0	0
71	60	4	7	-3	9
62	65	6	5	1	1
64	56	5	9	-4	16
58	59	8	8	0	0
80	77	1	1	0	0

Eng	Maths	R _x	R _y	d=R _x -R _y	d ²
76	67	2	3	-1	1
61	63	7	6	1	1
					Σd ² =54

$$\rho = 1 - (6 \times 54) / (10^3 - 10) = 1 - 324 / 990 = 1 - 0.3273 = 0.6727$$

$$\therefore \rho = 0.6727$$

SOLVED PROBLEM 4 — REPAIR JOBS (ESTIMATED VS ACTUAL COST)

A random sample of repair jobs: compute rank correlation.

Given Data (n = 8)

Estimated (x)	300	450	800	250	500	975	475	400
Actual (y)	273	486	734	297	631	872	396	457

Table 10.4

x	y	R _x	R _y	d	d ²
300	273	7	8	-1	1
450	486	5	4	1	1
800	734	2	2	0	0
250	297	8	7	1	1
500	631	3	3	0	0
975	872	1	1	0	0
475	396	4	6	-2	4
400	457	6	5	1	1
					Σd ² =8

$$\rho = 1 - (6 \times 8) / (8^3 - 8) = 1 - 48 / 504 = 1 - 0.095 = 0.9047$$

$$\therefore \rho = 0.9047 \text{ (Very strong positive rank correlation)}$$

SOLVED PROBLEM 5 — THREE JUDGES (MUSIC CONTEST)

10 competitors in a music contest are ranked by 3 judges A, B, C. Find which pair of judges have the nearest approach to common taste.

Rankings by 3 Judges

Comp	1	2	3	4	5	6	7	8	9	10
A	1	6	5	10	3	2	4	9	7	8
B	3	5	8	4	7	10	2	1	6	9
C	6	4	9	8	1	2	3	10	5	7

Table 10.5: Three Judge Analysis

A	B	C	dAB	dBC	dCA	d ² AB	d ² BC	d ² CA
1	3	6	-2	-3	5	4	9	25
6	5	4	1	1	-2	1	1	4
5	8	9	-3	-1	4	9	1	16
10	4	8	6	-4	-2	36	16	4
3	7	1	-4	6	-2	16	36	4
2	10	2	-8	8	0	64	64	0
4	2	3	2	-1	-1	4	1	1
9	1	10	8	-9	1	64	81	1
7	6	5	1	1	-2	1	1	4
8	9	7	-1	2	-1	1	4	1
						Σ=200	Σ=214	Σ=60

$$\rho_{AB} = 1 - 6(200)/(1000 - 10) = 1 - 1200/990 = -0.21$$

$$\rho_{BC} = 1 - 6(214)/990 = 1 - 1284/990 = -0.297$$

$$\rho_{CA} = 1 - 6(60)/990 = 1 - 360/990 = 0.636$$

KEY POINT: $\rho_{CA} = 0.636$ is positive and closest to +1. Therefore, Judges C and A have the nearest approach to common taste in music.

11. IMPORTANT FORMULAS SUMMARY

vtu-easy.com

Curve Fitting Formulas

Table 11.1: Curve Fitting Summary

Topic	Formula / Key Equation
Straight Line ($y=ax+b$)	Normal Eqns: $\Sigma y=a\Sigma x+nb$; $\Sigma xy=a\Sigma x^2+b\Sigma x$
Straight Line ($y=a+bx$)	Normal Eqns: $\Sigma y=na+b\Sigma x$; $\Sigma xy=a\Sigma x+b\Sigma x^2$
Parabola ($y=ax^2+bx+c$)	Normal Eqns: $\Sigma y=a\Sigma x^2+b\Sigma x+nc$; $\Sigma xy=a\Sigma x^3+b\Sigma x^2+c\Sigma x$; $\Sigma x^2y=a\Sigma x^4+b\Sigma x^3+c\Sigma x^2$
Power Curve ($y=ax^b$)	$\log y = \log a + b \log x \rightarrow Y = A + BX$; $a = e^A$, $b = B$

Correlation Formulas

$$r = \Sigma XY / (\sqrt{\Sigma X^2} \cdot \sqrt{\Sigma Y^2}) \text{ where } X=x-\bar{x}, Y=y-\bar{y}$$

$$\sigma_x = \sqrt{(\Sigma X^2/n)}, \sigma_y = \sqrt{(\Sigma Y^2/n)}$$

$$-1 \leq r \leq +1$$

Regression Formulas

$$\text{Line of } y \text{ on } x: y - \bar{y} = byx(x - \bar{x}), byx = r \cdot (\sigma_y / \sigma_x)$$

$$\text{Line of } x \text{ on } y: x - \bar{x} = bxy(y - \bar{y}), bxy = r \cdot (\sigma_x / \sigma_y)$$

$$r = \pm \sqrt{byx \times bxy}$$

Angle Between Regression Lines

$$\tan \theta = (\sigma_x \cdot \sigma_y / (\sigma_x^2 + \sigma_y^2)) \cdot ((1-r^2)/r)$$

Standard Error of Estimate

$$S_{yx} = \sigma_y \cdot \sqrt{(1-r^2)}, S_{xy} = \sigma_x \cdot \sqrt{(1-r^2)}$$

Spearman's Rank Correlation

$$\rho = 1 - (6\Sigma d^2)/(n^3-n) \text{ where } d = R_x - R_y$$

Table 11.2: Quick Selection Guide

Formula	When to Use	Key Sums Required
$y = ax + b$ (or $a+bx$)	Linear data trend	$\Sigma x, \Sigma y, \Sigma xy, \Sigma x^2$
$y = ax^2+bx+c$ (or $a+bx+cx^2$)	Curved data (quadratic)	$\Sigma x, \Sigma y, \Sigma x^2, \Sigma x^3, \Sigma x^4, \Sigma xy, \Sigma x^2y$
$y = ax^b$	Power law data (log-log linear)	$\Sigma(\ln x), \Sigma(\ln y), \Sigma(\ln x)^2, \Sigma(\ln x)(\ln y)$
Pearson r	Both variables quantitative, linear	$\Sigma X, \Sigma Y, \Sigma XY, \Sigma X^2, \Sigma Y^2$
Spearman ρ	Ranked data or ordinal	Σd^2 where d = rank difference

All the best for your VTU Examinations! — vtu-easy.com

vtu-easy.com | Curve Fitting, Correlation & Regression | Module 5